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Sheet 1

NAME Nour Ehab Samir

ID *20241063

Problem #1

$$a) 00010000_2 \xrightarrow{C1} \begin{pmatrix} 11101111 \\ 11110000 \end{pmatrix}_1 \xrightarrow[C2]{+1} (11100000)_2$$

$$b) 00000000_2 \xrightarrow{C1} (11111111)_1 \xrightarrow[C2]{+1} \cancel{(1000000000)}_2$$

$$c) 11011010_2 \xrightarrow{C1} (00100101)_1 \xrightarrow[C2]{+1} (00100110)_2$$

$$d) 10101010_2 \xrightarrow{C1} (01010101)_1 \xrightarrow[C2]{+1} (01010110)_2$$

$$e) 10000101_2 \xrightarrow{C1} (01111010)_1 \xrightarrow[C2]{+1} (01111011)_2$$

$$f) 11111111_2 \xrightarrow{C1} (00000000)_1 \xrightarrow[C2]{+1} (00000001)_2$$

Problem #2

122	0
61	1
30	0
15	1
7	1
3	1
1	1
0	

$$(11000100)_2$$

$$(1111010)_2$$

$$(11000100)_2$$

$$(11000100)_2$$

$$(11000100)_2$$

$$a) 11111010_2$$

MSB = 1

$$b) (01111010)_2 \xrightarrow{C1} (10000101)_1 \xrightarrow[C2]{+1} (10000110)_2$$

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problem #3

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42	0	A	77	1	A
21	1		38	0	
10	0		19	1	
5	1		9	1	
2	0		4	0	
1	1		2	0	
0			1	1	
			0		

a) $(10101010)_2 \rightarrow (-42)$
 $\hookrightarrow$ MSB-1

$(00101010)_2 \xrightarrow{+1} (11010101)_2 \xrightarrow{+1} (11010110)_2$

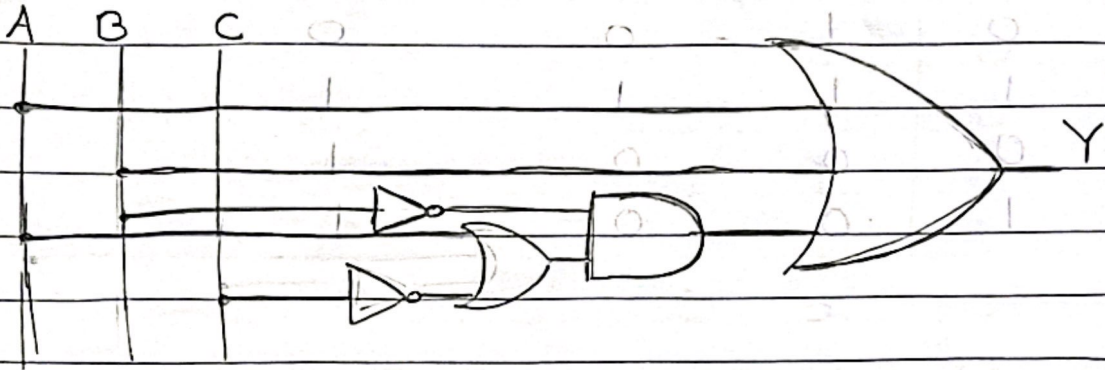
b) $\begin{array}{r} \times \quad 01001101 \\ + \quad 11010110 \\ \hline \end{array}$ (77)
 (-42)

$\text{+ve no.} \leftarrow (0100011)_2$
 $(00100011)_2$

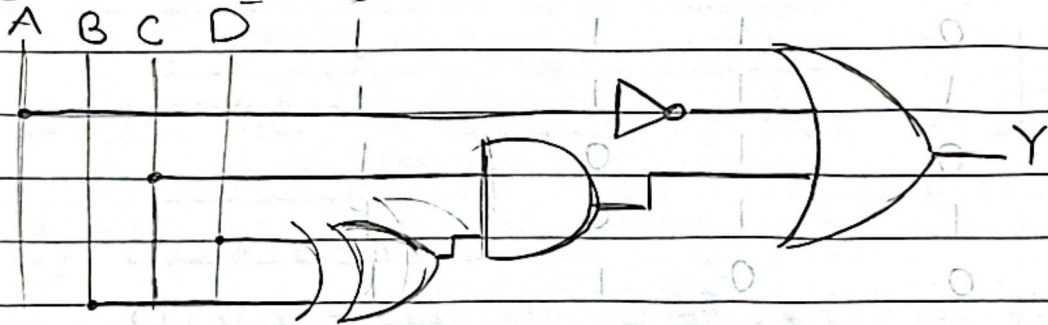
$\Downarrow$
 $(35)_{10}$

Problem #4

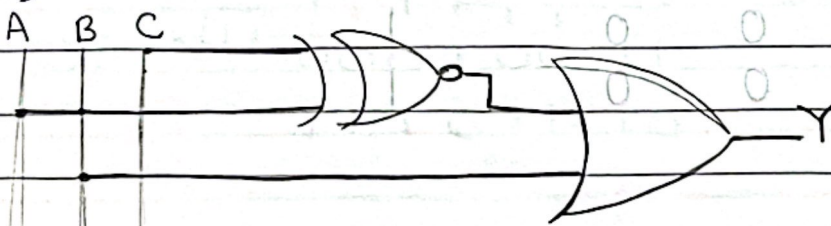
a) $Y = A + B + B'(A + C)$



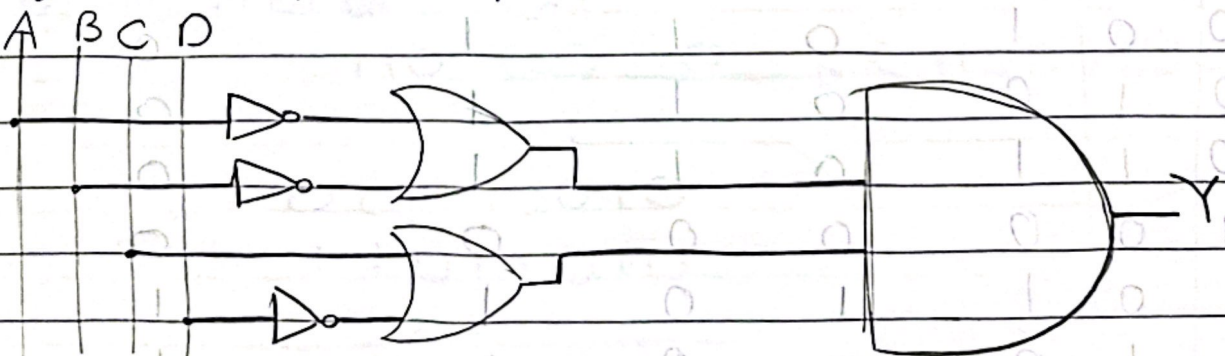
b) $Y = C(C \oplus D) + A' A$



c) $Y = (A \oplus C)' + B$



d) $Y = (A' + B')(C + D')$



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problem #5

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a)	A	B	A'	A'.B	(A + A'.B) = Y
	0	0	1	0	0
	0	1	1	1	1
	1	0	0	0	1
	1	1	0	0	1

b)	A	B	C	A'	B'	A + B' + C = Y
	0	0	0	1	1	1
	0	0	1	1	1	1
	0	1	0	1	0	1
	0	1	1	1	0	1
	1	0	0	0	1	1
	1	0	1	0	1	1
	1	1	0	0	0	1
	1	1	1	0	0	1

c)	A	B	C	AC	A'	(B + AC + A') = Y
	0	0	0	0	1	1
	0	0	1	0	1	1
	0	1	0	0	1	1
	0	1	1	0	1	1
	1	0	0	0	0	0
	1	0	1	1	0	1
	1	1	0	0	0	1
	1	1	1	1	0	1

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d)

A	B	C	B'	AB	AB'	B'C	(AB + AB' + B'C)
0	0	0	1	0	0	0	0
0	0	1	1	0	0	1	1
0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	0
1	0	0	1	0	1	0	1
1	0	1	1	0	0	1	1
1	1	0	0	1	0	0	1
1	1	1	0	1	0	0	1

Problem #6

a) 00010011 (19)

00010010 (18)

C2

(11101110)C2

(-18)

ignore
carry

11111111
00010011

+ 11101110

00000001 → (1)₁₀

b) 00100010 (34)

00100110 (38)

C2

(11011010)C2

(-38)

00100010

+ 11011010

11111000

MSB = 1

(-ve)

(C1 → 00000011 + 1 → 00000100

2's Comp
القيمة العكسية

(-4)₁₀

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c) $00001001_{(9)} \xrightarrow{C_2} 00110101_{(53)} \xrightarrow{C_2} (11001011)_{C_2}$

00001001
 11001011

MSB-1 $\leftarrow$ 11010100

$(C_1 \rightarrow 00101011 + 1 \rightarrow 00101100$

$\downarrow$

$(-44)_{10}$

d) 00101000 (40) $\xrightarrow{(-21)}$ 00010101 (21) $- C_2 \rightarrow$ (11101011) C_2

ignore carry $\rightarrow$ $\begin{array}{r} \times 111 \\ 00101000 \\ + 11101011 \\ \hline 00010011 \end{array}$ $\rightarrow (19)_{10}$

(1) 100000000

(38) 00100110 CS (39) 00100100 (24)

01000100

01011011

0 1 1 1 1 1 1